Student Engineer Handbook

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1. Introduction

Praxis is a first year and second year Engineering Science (EngSci) course that introduces students to the foundation of EngSci's design education. This course introduces the models and tools of engineering design, communication, and teamwork. It helps students develop various skills and knowledge associated with approaching an engineering design and engineering practice, and it also provides the opportunity for students to advance their teamwork and communication skills through interaction with the public.

This handbook is intended to serve as a reference to provide guidance for future engineering design activities for individuals who are already familiar with the topic area, such as the author, engineers, and engineering students. It draws upon the experiences of the author, Eric Miao, a first year Engineering Science student at the University of Toronto.

The handbook uses concise language and is structured for easy navigation by utilizing numbered sections and page numbers. To identify subsubsections easier (sections with four or more numbers), they will be bolded. As one of the stakeholders is the author, this handbook is written in a first-person narrative.

This handbook will present the following major topics:

- My positionality and values as an engineering student, to allow the readers to understand the potential biases I may have.
- The engineering design process that I use when I approach an opportunity.
- The specific tools, models, and frameworks I have used as a part of the engineering design process, and specific supporting evidence(s) for each.

2. Positionality and Values

This section focuses on my positionality as an engineering student. Furthermore, it discusses my values when I approach an engineering design process. This section will allow the reader to understand more about myself and my potential biases with this handbook, the engineering design, and the engineering design process. To begin, my positionality and how they may contribute to my biases in this handbook are discussed. After, my values and how they may contribute to the biases in this handbook are discussed.

2.1. Positionality and Biases

I believe engineering is a field that either designs something new or improves on a design or concept, with the intention of making certain tasks easier for the public.

I identify myself with three key roles: an engineering student, an educator, and a leader.

2.1.1. Engineering Student

As an engineering student, I enjoy learning and always seeking for applications of the materials I learned to enhance my understanding of the materials. This role may contribute to my bias, as I recognize being a student, I am still new to the formal engineering processes and may not have full understanding of the various tools in this handbook.

2.1.2. Educator

As an educator, I enjoy teaching others interesting material that they can apply later in life. As an instructor at Engineering for Kids, I get first-hand experience of the excitement in STEM classrooms of various aged students, and what teaching STEM courses is like. This may contribute to my bias as this role is to expect students to apply the materials that I taught, so as a student I may not fully understand the purpose of the tools in this handbook and the various ways to use these tools.

2.1.3. Leader

As a leader, I enjoy bringing everyone together to achieve specific goals and seeing success. During school projects and teams outside of school, I learned that being a leader is not only to make projects familiar to the team and distribute work, but also to learn

specific skills everyone possesses and how these skills can be used in the project. The bias this role may contribute to is like the one in educator role.

2.2. Values

When approaching an engineering design process, I generally have these core values: design for quality and functionality, design for aesthetics, and design for durability and sustainability. These three values are similar of those in my position statement four months ago, but there are some evolvements. In addition, these values are likely to change based on the nature of the project. For instance, I value design for simplicity more than my main values in one of my projects.

Thus, due to the constant evolving of my values towards engineering designs, this handbook will have biases towards and focus more on the three main values I listed above, that might differ from my future self and other engineers.

Outside of academic environment, I have three main values: accountability, organization, and elation.

- Accountability – I do not like making promises without fulfilling them. However, some situations can be unforeseen. If those situations are likely to occur, I will give early heads-up or as soon as I make aware of the situations. Being accountable also means to me that I should be aware of situations when others are unaware, such as the safety of friends and family always.
- Organization – During my teenage time, I discover that I am highly organized in various ways. During hang outs, I am aware of the different aspects and assume accountabilities for all those aspects, such as purchasing history made by friends for friends, organizing location and time of hang outs. For general organization, I always organize everything in the best way possible, whether that's files and holders on computers or shelves at home.
- Elation – No matter where I go or who I'm with, I always ensure the joy of others. I believe I have a sense of others mood, and I always assume accountability to make others happy, whether by making them known of their existence or just chatting with them.

3. Engineering Design Process

The engineering design process is a fundamental process when approaching any engineering problems. This process may differ industry by industry, and individual by individual. *Figure 1* is a model of the engineering design process that I use, and it can be loosely categorized using the FDCR braid model.

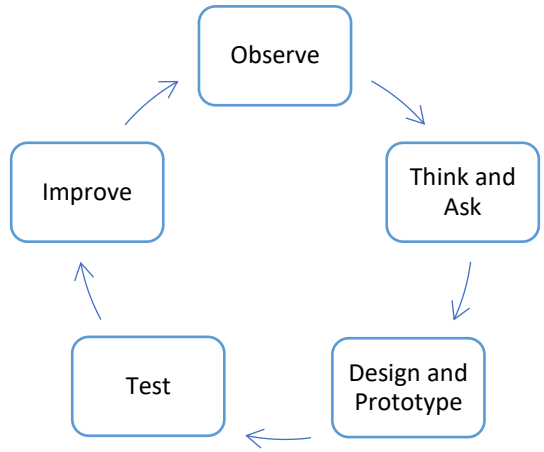

Figure 1 Engineering Design Process Cycle

3.1. Observe

Observation is my first step of the engineering design process. Before I proceed further into the process, I must know about the problem by observing. This phase can be categorized as the beginning phase of *framing* of the FDCR model.

3.2. Think and Ask

After the observation, I must think about the problem and ask for details about it. This may include the objectives of the design, aspects of the design, and implementation of the design. This phase can be categorized as the rest of *framing* of the FDCR model.

3.3. Design and Prototype

Indicated by its name, this phase is to design a solution and prototyping it. This phase of the engineering design process is the combination of *diverging*, *converging*, and *representing* of the FDCR model.

3.4. Test

After making a prototype, this step is to test the design to answer the questions and meet the objectives from 3.2.. This step belongs to the *represent* of the FDCR model.

3.5. Improve

During testing phase (3.4.) if it does not satisfy the objectives or answer the problems, this step is, as indicated, to improve the design and/or prototype. To do this, I must observe (3.1.) what goes wrong or what the problem is and think (3.2.) about a solution for it, which then goes through the entire cycle of the process again.

4. Tools, Models, and Frameworks

4.1. Framing

Framing refers to understanding the situation. This step defines an engineering opportunity and requirements with the stakeholders.

4.1.1. Primary and Secondary Research

Research is to find sources for proof, information, processes, and related concepts. The motivation for research is to discover something new through testing or online research. Primary research is mainly an interaction with stakeholders or testing and observations. Secondary research is mainly finding articles, research papers, standards, codes, or existing designs related to the problem that may guide the designs.

4.1.2. Requirements Model

The purpose of the requirements model is to outline the requirements that a design should or must achieve. It is interpreted from stakeholders' statements.

The requirements model should consist of higher objectives, which is to outline the general idea of what the design must achieve. Within the higher objectives, it consists of several detailed objectives described to support the validity of the higher objective. Each objective should contain its associated metrics, criteria, and constraints. Metrics is defined by the unit the objective measures; criteria is defined by the gradient of the metric (higher or lower); constraints is defined by the cut-off of the criteria and is normally presented as a numerical value with justification.

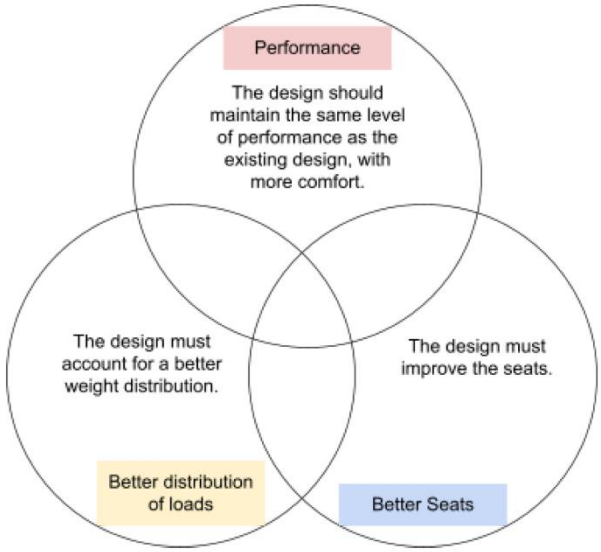

Figure 2 high level objectives for police cruiser RfP [3]

4.1.2.1. Example

Figure 2 is one example of the high-level objectives [3]. It outlines what our problem addresses generally, and specific requirements revolves around these three objectives. This diagram complies with the team values of being concise and simple, and it is easy to understand for the reader.

Although this diagram gives a clear idea about the three high level objectives, but it lacks descriptions of the overlapping parts, which results in a misunderstanding of why this model is represented in Venn diagram or what the diagram represents.

4.1.3. CRAAP Test

When performing secondary research, there are going to be lots of information. The CRAAP test is useful to evaluate the credibility of the information. It helps students evaluate the information based on five different criteria:

- Currency: the timeliness of the information.
- Relevance: the importance of the information for your needs.
- Authority: the source of the information (author, organization, qualification, etc.).
- Accuracy: the reliability, truthfulness, and correctness of the content.
- Purpose: the reason the information exists.

4.1.4. Reframing, Rescoping, and Refining

The purpose of reframing is to modify the situation to address another problem given the stakeholders statement. The purpose of rescoping is to modify the target of interest within the frame of the problem. The purpose of refining is to improve clarity, detail, and substantiation of the problem within the frame and scope of the problem. Both reframing and rescoping may potentially result in changing the requirements model.

Reframing or rescoping should be done if the diverging process is too restricting. It is useful to broaden the criteria, allowing a more variety of ideas. Refining is done by clarifying with the stakeholders about any uncertainties in the problems.

4.1.4.1. Example

- **Reframing**: We considered the only stakeholder to be students
  - profs were not considered to be a relevant stakeholder since students untangling a cable does not take away from a professor's lecture time
  - however, it may affect a class's overall performance in a course since a majority of the students are spending time untangling cords instead of paying attention
  - This may affect the professor by affecting their ability to teach a course effectively but it is likely by an insignificant amount.

Figure 3 reframing of the design brief in Praxis I [5]

Figure 3 shows an example of reframing of the organizing charging cables design brief in Praxis I [5]. This is considered a reframing as a change in stakeholder would result in a change of the nature of the situation and the target of the design, even if the scope remains the same. As one of the team values is simplicity, this reframing is done to simplify the opportunity.

4.2. Diverging

Diverging refers to coming up with lots of ideas. This step generates ideas and exploring alternatives using various methods.

4.2.1. Classical Brainstorming

Classical brainstorming is giving the problem and allowing every individual to come up with ideas themselves. This is a perfect tool to be inclusive, as this is a group diverging tool and everyone will be able to participate in the generation of ideas. This tool is also beneficial in learning the bias each person has and addressing these biases after the divergence is also important. Procedure:

- Introduce the problem to everyone, make sure everyone understands the problem
- During the idea generation process,
  - make sure no criticisms or judgements are made (everyone stay quiet except for the one saying the idea)
  - only one person speaks at a time and must be relevant to the idea generation
- Combine similar or identical ideas
- Get a consensus of which ideas may be investigated further

4.2.1.1. Example

Figure 4 shows the result of classical brainstorming for protecting telescope [4]. By doing classical brainstorming as our first divergence tool, it allows our team to realize that everyone has a common cognitive bias for a big covering over the telescope, rather than something more creative. Since everyone has a common bias for this problem, this method does not work well with us and therefore not a lot of ideas are generated.

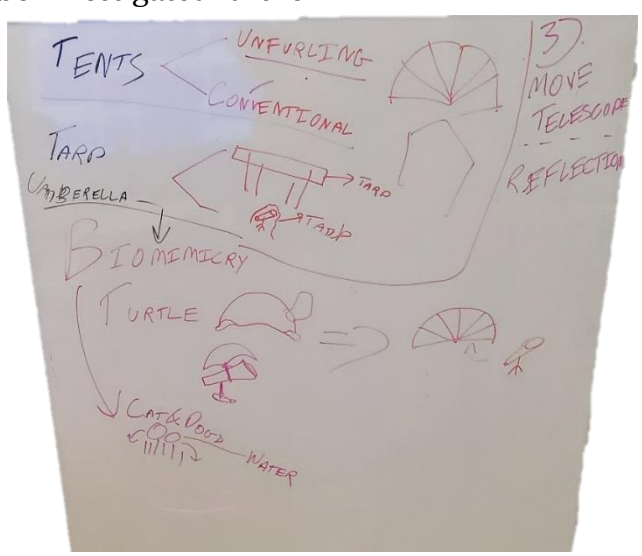

Figure 4 ideas written during classical brainstorming for protecting telescope [4]

4.2.2. Reverse Brainstorming

Reverse brainstorming does not address the problem, instead it seeks for ways to cause the problem. This way, the cause of the problem is acknowledged, and a solution may be available to address it. This tool is beneficial to address any biases against solutions focusing on one cause, where a problem may have multiple causes. In addition, this tool

can be used for independent brainstorming or a group brainstorming. Thus, this tool can also be used for an inclusive environment. Procedure:

- Identify the problem and its potential cause(s)
- Identify ways to prevent the cause(s) from happening, either a known solution or a design to prevent the problem being caused

This method is extremely useful and is usually my first method of diverging as I seek for a design for functionality. By addressing the cause of the problem, the design would theoretically be functional.

4.2.2.1. Example

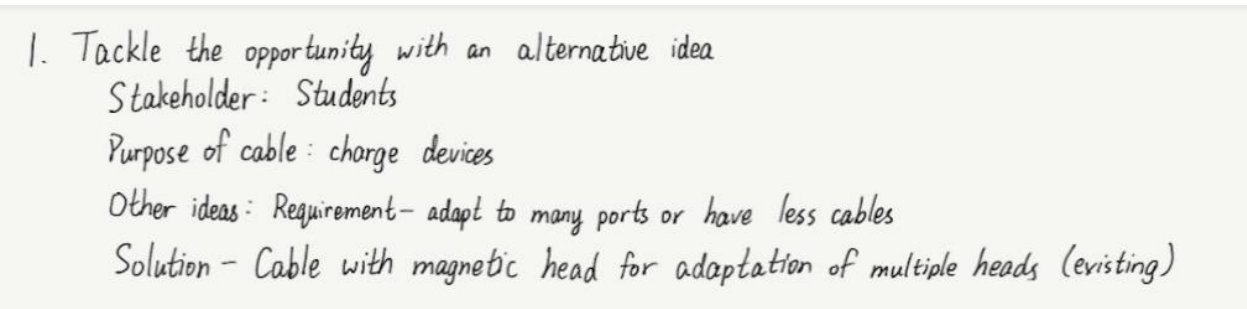

Figure 5 my thought process of untangling cables [5] using reverse brainstorming

Many of my team members choose to tackle the problem directly, but *figure 5* shows my thought process of tackling the problem from its cause. In charging cables design [5], I identify that having devices with different ports may be the cause, or the different lengths of the cables can be a cause. Thus, to tackle both causes for this problem, I implement an exchangeable port by having a magnetic connection head and an extendable wire from a small container so the users can adjust the length to their preference. Again, I have an extreme bias for this method, but it works very well regardless of what the problem is.

4.2.3. Attribute Listing

Attribute listing is breaking the problem down into smaller bits and pieces. This will allow all aspects of the problem to be examined. This method can be used for individual divergence or a group divergence. It is a useful tool for inclusiveness, as everyone can participate in the breaking down process and the examination process. Furthermore, it can be used to avoid any biases against anchoring. Procedure:

- List as many attributes of the object in question
- Evaluate the values of the attributes
- Modify the attributes to create values and address the problem

4.2.3.1. Example

Fabric Material	Vinyl Fabric (durable, flexible, versatile)	Polyester Fabric (more durable than vinyl, can also withstand oil and chemicals)	Polyethylene Woven Fabric (less tear-resistant than vinyl, high tensile strength)

Figure 6 section of an attribute listing showcasing the process for protecting telescope [4]

Attribute listing requires an object or a design concept to be broken down. Shown in *figure 6*, one of the attributes of the design for protecting telescope is the hydrophobic fabric material that the design will use. By breaking down to parts, it allows us to examine the parts more closely, and allows us to understand the situation better. This method works well with protecting telescope, as the motivation of the problem is protection against precipitation, so breaking down the design aspects allows us to modify one part at a time to ensure the effectiveness of the design.

4.2.4. Biomimicry

Biomimicry is a diverging method where inspirations come from the study of nature. These inspirations may come from how an animal may possess the function to solve a problem, or how a plant may possess the quality of the problem. Procedure:

- Identify key elements of the problem
- Research or think about if nature replicates a similar situation
- If such information exists, research or think about if nature possess a solution for the situation and discuss if this can be implemented as a solution

4.2.4.1. Example

Figure 8 Chameleon's tongue rolling inspiration [1] [5]

Figure 7 real life design inspired by Chameleon [5]

One of the designs for organizing charging cables is a recoiling children's wristband. This idea originally comes from inspecting a Chameleon's tongue in action, and an inspiration for rolling to store things comes up. Then, by inspecting real life objects related to rolling

to expand and recoil, we figure the wristband is perfect for this design concept. Biomimicry may work for some designs but may not for others. In this case, it is the perfect tool to generate a creative solution, and everyone likes it.

4.2.5. Brainwriting 6-3-5

Brainwriting 6-3-5 is a group diverging method. It allows for individuals to come up with solutions and allows for peer editing. This method is the best for being inclusive and for avoiding bias. It is an inclusive method because everyone can generate ideas, and it is an anti-bias method because everyone can peer edit all the solutions and give comments to address biases a solution may have. Procedure:

- Establish a team of 4-8, generally 6 people
- Each member writes down 3 ideas, make sure there are no discussion
- Pass the concepts to the next person, and each member will revise the three ideas that the previous person wrote
- Repeat 5 times

4.2.5.1. Example

Figure 9 below shows our team's process for the protecting telescope [4]. Each one of our three members have different colour markers to show the process. This method is used by our team as we want to break free from our cognitive bias against a big tarp covering the telescope, so one condition is to avoid that type of idea during our 6-3-5 process. This tool is very helpful in terms of the idea generation, but it does extremely well with our purpose of using the tool, which is to break free from our anchoring. As shown, without covering the telescope, our team is able to come up with seven different solutions to tackle the problem.

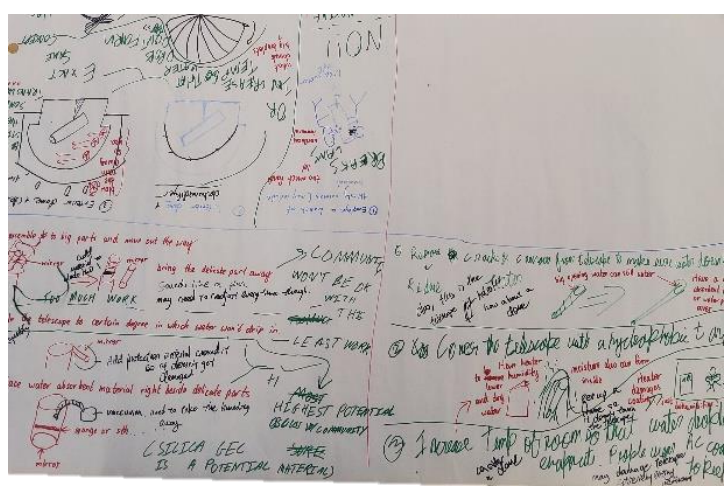

Figure 9 6-3-5 method for protecting telescope [4]

4.2.6. Morphological Chart

Morph chart, like attribute listing, breaks the problem down to smaller bits and pieces, usually as aspects or features of the design. Each aspect or feature of the design is accompanied by a number of designs related to it. This is then put into a table, and the motivation for morph chart is to explore the different combinations of the features. Procedure:

- List as many attributes (aspects or features) of the design
- List as many designs related to each attribute and put them in a table
- Explore the different possible combinations

4.2.6.1. Example

Our team used a morph chart to see what combinations of designs would result with the best look and functionality as those were two of my values. Shown in *figure 10*, it lists the aspects of the design for the solution. With each aspect, we list out every possible solution to maximize the possible combinations. Although it works well for my design's purpose to make it look good, it would require a layout of a design or an object that already exists to solve the problem. Therefore, this tool may not be useful to commence the divergence process.

Purpose
Waterproof material
Location respect to the telescope in which the waterproof material should be placed
Shape/Struct of the material

Figure 10 section of a morph chart for protecting telescope [4]

4.3. Converging

Converging refers to throwing some ideas away. This step requires making decisions and justifying recommendations.

Often these tools for convergence rely on the key objectives in the objective model. Thus, the authoring team should

- identify the key objectives based on their values, and
- retrieve the metrics, criteria, and constraints of the key objectives.

4.3.1. Proxy Testing

Proxy testing requires either research or prototypes for physical testing. The motivation for proxy testing is to test a specific aspect of the requirements model for all the designs. The values for the metrics can either be obtained by conducting research or by physical testing. Procedure:

- Choose an objective from the requirements model (key objective preferred)
- Obtain its associated metric(s) and evaluate how it can be measured
- Conduct research or testing to obtain the measurement of the metric(s)

4.3.2. Measurement Matrix

A measurement matrix is the most generic tool for convergence, and most effective tool for convergence. It compares all designs in all metrics with measurements obtained by conducting proxy testing (4.3.1.). Procedure:

- Create a table with metrics for each column and designs for each row
- Input the measurement of metrics associated with each design

4.3.2.1. Example

Measurements Matrix	
COST ($)	15
MASS (gram)	80
VOLUME (cm^3)	45
SET-UP TIME	Low
TIME (sec)	9.9
ADAPTABILITY (# of Cables)	1

Figure 11 section of a measurement matrix for organizing cables [5]

Figure 11 shows a section of the measurement matrix for the organizing cables [5] spartz. This specific section is chosen because it shows the metrics of all the objectives, and their quantitative measurements for a specific design. This tool is very effective since all the measurements are laid out for comparison, and other convergence tools may use the measurements from this tool.

4.3.3. Pairwise Comparison Matrices

This converging method usually utilizes the data obtained from the measurement matrix (4.3.2.). It chooses a specific objective and its associated metric(s) and compare a pair of designs based on it. This method is useful for estimating which design would be the best for the objective but cannot by justified just by this converging method. Procedure:

- Create a square table
- Input metric to top left box
- Make the rest of the rows and columns to each design

- Diagonals are marked with ~ since those boxes are comparing each design with itself
- For each box, compare the design associated with the row with the design associated with the column. If the design in row is better than the design in column, input 1 or + in the box, otherwise 0 or –

4.3.3.1. Example

Number of Water Drops	Patching Dome holes w/ tape	Fan system	inflating balloon	Detachable umbrella	Folding Dome	Hydrophobic Coating
Patching Dome holes w/ tape	-	-1	-1	-1	-1	0
Fan system	1	-	-1	-1	-1	1
inflating balloon	1	1	-	1	-1	1
Detachable umbrella	1	1	-1	-	-1	1
Folding Dome	1	1	1	1	-	1
Hydrophobic Coating	0	-1	-1	-1	-1	-

Figure 12 a pairwise comparison matrix for number of water drops for protecting telescope [4]

In *figure 12*, a pairwise comparison is done with respect to the number of waterdrops for the opportunity of protecting the telescope [4]. We present the values numerically as we want to sum each row and compare the sums to estimate which design is the best concept with respect to this particular metric. This allows us to better understand which concept has potential to be the final design, but we acknowledge that we cannot justify anything only by using a pairwise comparison matrix. However, this graph may be really confusing to others, even within the team. I have to explain what the columns, rows, and numbers mean. Therefore, if a team does not reach consensus about how to interpret a graph, pairwise comparison matrix is not recommended.

4.3.4. Pugh Chart

A Pugh chart uses a reference design and compare it with the other designs in all metrics. This method should be used if a problem has an already-existing design. It is useful when comparing new solutions with an existing solution. Since engineering is about improving something, this method is reliable as it can determine whether one should continue with a design or resort back to the existing design based on the results. Procedure:

COST	-
MASS	-
VOLUME	+
SET-UP TIME	o
TIME	-
ADAPTABILITY	o

Figure 13 section of a Pugh chart for organizing cables [5]

- Set up a table with its columns as different designs and rows as different metrics
- Choose a pivot (reference design) and mark all its metrics to 0
- Compare other designs with the pivot in each metrics, and mark the box with 0 if the design is the same as the pivot, + if the design is better than the pivot or – otherwise

4.3.4.1. Example

A Pugh chart also uses the measurements obtained from the measurement matrix (4.3.2.). In *figure 13*, we choose our reference as the design that is our preferred concept, then we compare a design with our reference design in all metrics. Using this tool allows us to understand whether another design is better performed than what we prefer. I prefer Pugh chart because not only is it easier to read and understand for myself, but also for others. Furthermore, Pugh chart compares all possible metrics whereas a pairwise comparison only does one, so Pugh chart is more efficient.

4.4. Representing

Framing, diverging, and converging all needs to be represented in some way. There are various ways to represent. The key here is to show a key aspect of a design; each piece must have a meaning within them. Representing is usually done with various types of prototypes. They are most commonly functional prototype, graphical prototype, mathematical prototype, and simulation prototype. The purpose of representing is to show what one learns from it or to justify an aspect of the design.

4.4.1. Functional Prototype

A functional prototype means a physical prototype for the purpose of testing or showcasing. This type of prototype can physically showcase a specific feature and/or a functionality of a design. A functional prototype can also come in the shape of a photograph or a video recording of the functionality of a design.

4.4.1.1. Example

In *figure 14*, the aspect of this prototype is to show how the design would function, and to enhance one's understanding of the concept for

Figure 14 a physical prototype for protecting telescope [4] showcasing the functionality of the design

the problem. When making a functional prototype, make sure that the prototype has a meaning to it.

4.4.2. Graphical Prototype

A graphical prototype is a prototype showcased using a sketch or other graphical representations. This type of prototype is best for relaying theoretical features of a design. This type of prototype is useful when a physical prototype to showcase the features is not feasible. When making a graphical prototype, ensure that it has meaning to it and others can easily interpret what the prototype does.

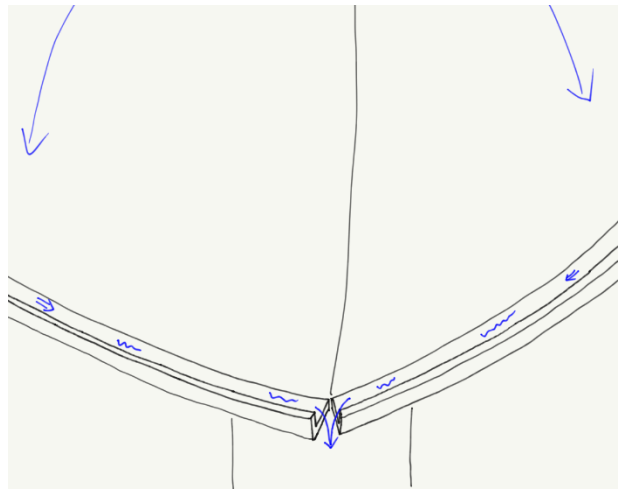

Figure 15 a theoretical representation of water tracks for protecting telescope [4]

4.4.2.1. Example

Figure 15 shows the graphical prototype of an optional feature for a design in protecting the telescope. A water track can guide fallen water towards a specific point of the design to allow it to drain. As my team does not have enough time to prototype physically, we instead use a graphical prototype to showcase this feature.

4.4.3. Mathematical Prototype

A mathematical prototype is a prototype showing mathematical aspects or features of the design. It can relate specifically to the dimensions of a concept, the geometry, or also the properties of materials. A mathematical prototype is usually used to verify the functionality and feasibility of a design, but can also be used for decision making.

4.4.3.1. Example

Figure 16 shows a calculation of the distance between the theoretical tarp from the telescope body for protecting telescope [4]. It uses simple geometry, trigonometry, and algebra. This prototype helps us verify the feasibility of this design concept.

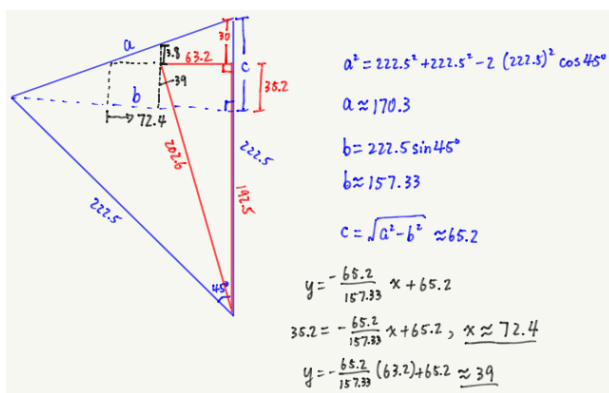

Figure 16 a mathematical model of protecting telescope [4] showcasing the distance between the design and the telescope to ensure telescope movement

4.4.4. Simulation Prototype

A simulation prototype uses computer programs to simulate a situation to showcase aspects or features of the design. This prototype is ideal for situations where physical testing cannot be done appropriately or safely. But it can also be used to showcase the impossible or intricate features of a design.

4.4.4.1. Example

In *figure 17*, it shows the beginning stage of our testing for water resistance. This is shown as a simulation not because the testing is inappropriate, but because not of the team members want to get wet. Afterall, this prototype helps us with verification. However, it does not consider the force water exerts on the fabric and the weight it can handle, and those should be done using a mathematical prototype.

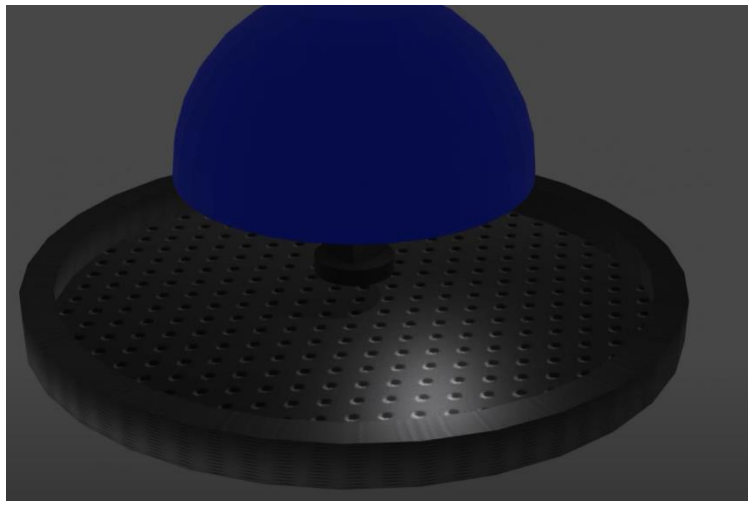

Figure 17 a simulation prototype showcasing the testing of water prevention for protecting telescope [2] [4]

5. Embedded Extract

[1] A. Bouchard. “How is it Possible for a Chameleon’s Tongue to Be So Quick?” labroots. 2016. Available: https://www.labroots.com/trending/plants-and-animals/2973/chameleon-s-tongue-quick

[2] A. Sachdeva. “Dome Covering on Telescope.” R4204: Praxis II. 2022. Available: https://www.youtube.com/watch?v=zz6thbqttcU

[3] R4204. “A Proposal to Improve Police Officer’s Sitting Comfort.” Praxis II. 2022.

[4] R4204. “Preventing Precipitation Ingress to and Damage of the Telescope at Allan I. Carswell Observatory.” Praxis II. 2022.

[5] WA4207. “Designing an Improved System for Organizing Charging Cables for University Students.” Praxis I. 2021.